E205 Lab 1 - Data Manipulation & Modeling

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***Abstract*— Files from E205_Lab1_data and police_killings.csv provided in the Lab prompt were used for this report. The data was obtained by setting the LiDAR in a stationary position in the Parsons building courtyard of Harvey Mudd College.**

I. Procedure

A. Creating Histograms

After downloading the data, a histogram was plotted using each of the following data vectors shown below:

Fig 1 - Range for Azimuth = -90 degrees

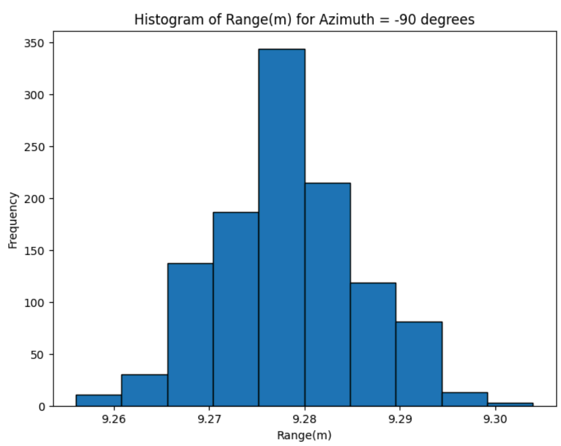


Fig.2 - Range for Azimuth = 0 degrees

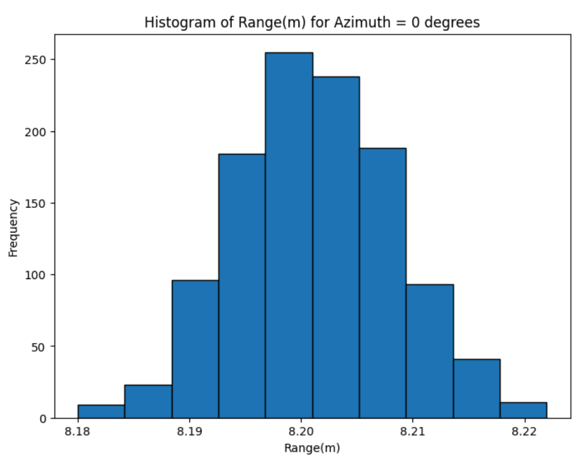
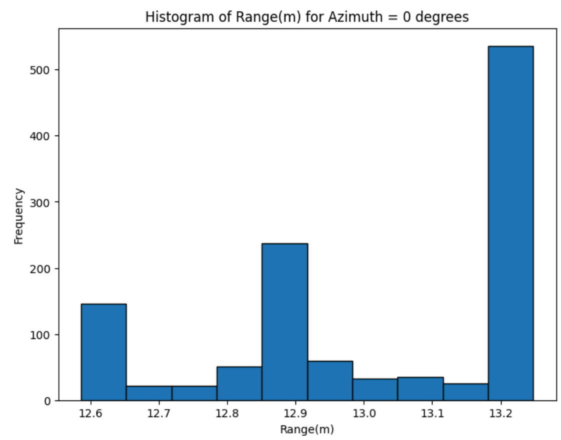


Fig.3 - Range for Azimuth = 90 degrees

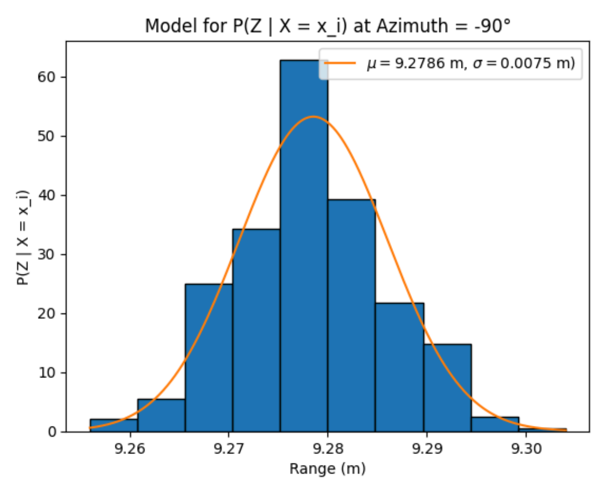


B. Creating Model

Next, the histogram in Fig.1 and azimuth data -90 degrees was used to create a mathematical function that best characterizes the data and can be used as a conditional probability function P(Z|X = x), where x is fixed since the sensor is stationary. We chose the Gaussian function for modeling, which requires finding the mean μ and the variance σ^2. The probability density function for the Gaussian Distribution is shown in eq.[1] and is implemented in Fig.4:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad [1]$$

Fig 4 - p(z|μ, σ^2) vs Range (m)



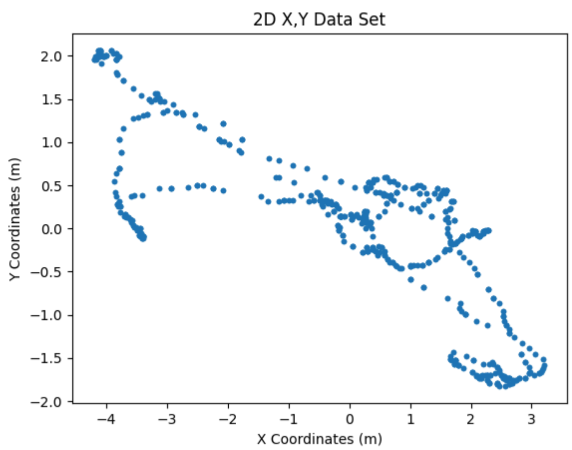
C. Transform and Plot GPS Measurements

Next, the GPS longitude and latitude measurements (from the azimuth=0 file) were transformed to X and Y coordinates in meters. The equations for transformation were found using E80 Lab 7 [2] and are labeled as Eq 2 and 3. The R_Earth constant used was 6335439m. We defined the origin to be located at the mean longitude and mean latitude measurements.

$$state.y \;=\; R\_earth * latitudeChange \qquad [2]$$

$$state.x \;=\; R\_earth * longituteChange * cos(latitude\ at\ origin) \qquad [3]$$

Fig. 5 - 2D X,Y GPS Measurements



The X-coordinate measurements exhibit a larger spread, and therefore higher variance, than the Y-coordinate measurements.

D. Implementing Bayes Rule

Next, we assume that the east and west walls are 22 m apart in the east west direction, so that the origin of the coordinate frame is now 11m distance to the nearest point on each wall. The closest point on the north wall is also 11m from the coordinate frame origin. We also assume that there are only four position states in which the sensor could be located. These are attached below as equation [4]

$$\begin{aligned} x_1 &= \begin{bmatrix} x_1 & y_1 \end{bmatrix} = [-1.70 \;\; 2.8] \\ x_2 &= \begin{bmatrix} x_2 & y_2 \end{bmatrix} = [-1.72 \;\; 2.8] \\ x_3 &= \begin{bmatrix} x_3 & y_3 \end{bmatrix} = [-1.74 \;\; 2.8] \\ x_4 &= \begin{bmatrix} x_4 & y_4 \end{bmatrix} = [-1.76 \;\; 2.8] \end{aligned} \qquad [4]$$

Bayes Rule – which relates a conditional probability to its inverse – was implemented and is referenced in equations [5] and [6] below.

$$p(x \mid y) = \frac{p(y \mid x)\,p(x)}{p(y)} \quad [5] \qquad p(x \mid y) = \eta\, p(y \mid x)\,p(x) \quad [6]$$

Equations [5] and [6] were then manipulated to give Equation [7]:

$$p(xi \mid z) \;=\; \eta \;*\; p(z \mid x) \;*\; p(x) \qquad [7]$$

The known values of p(xi) = 0.25, and z = 9.272m were used in Equation [7]. From then, we needed to find the normalization factor η, and $p(z \mid x)$. $p(z \mid x)$ can then be found using Eq [1] from the previous modeling of p(z|μ, σ^2) vs range (m) as seen in Fig.4. The variance σ^2 was assumed to have remained the same, and the mean μ was updated using the new coordinate frame, by taking 11m and subtracting the x coordinate from Eq [4] from it for each new $p(z \mid xi)$. This would give us $\mu_{x1} = 9.3$, $\mu_{x2} = 9.28$, $\mu_{x3} = 9.26$, $\mu_{x4} = 9.24$. Plugging each of these values into $p(z \mid x)$ gives us Table 1:

Table 1: $p(z \mid x)$ for each xi

$p(z \mid x1)$	$p(z \mid x2)$	$p(z \mid x3)$	$p(z \mid x4)$
0.049	30.11	14.78	0.0059

Table 2: $pbar(xi \mid z)$ for each xi

$pbar(x1 \mid z)$	$pbar(x2 \mid z)$	$pbar(x3 \mid z)$	$pbar(x4 \mid z)$
0.049 * 0.25 = 0.01225	30.11*0.25 = 7.5275	14.78 * 0.25 = 3.695	0.0059 * 0.25 = 0.001475

With Table 2 and Eq [7], we can find normalization factor:

$$\eta \;=\; \frac{1}{pbar(x1 \mid z) + pbar(x2 \mid z) + pbar(x3 \mid z) + pbar(x4 \mid z)} \;=\; 0.0889 \qquad [8]$$

Now we use Eqs [8] and [7] and plug in our known values to find p(xi| z) shown in Table 3 below:

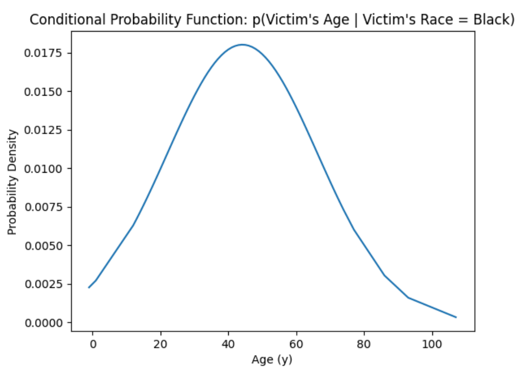
Table 3: p(xi| z) for each xi

p(x1| z)	p(x2| z)	p(x3| z)	p(x4| z)
0.00109	0.669	0.328	0.0001313

E. Creating Police Data Model

Last, a Police Data Model was made using police_killings.csv data set, which describes victims' name, age, race, sex, etc. The columns of the data set used for this model were age and race of the victim. In order to make this model, two methods were used. The first implemented the same Gaussian Distribution Eq [1], and the second involved selecting for a specific race within the age column of the data. The results of both are shown below:

Fig 6.1 a (method 1) above, b (method 2) below



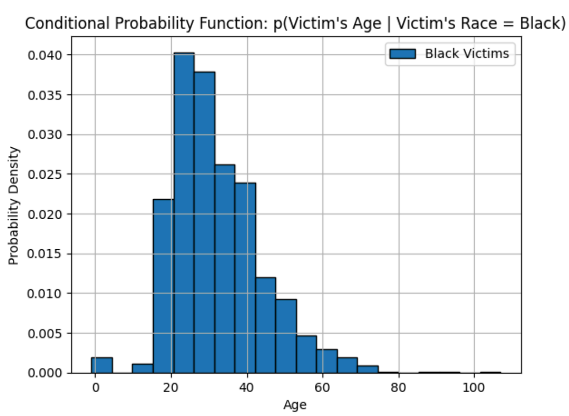
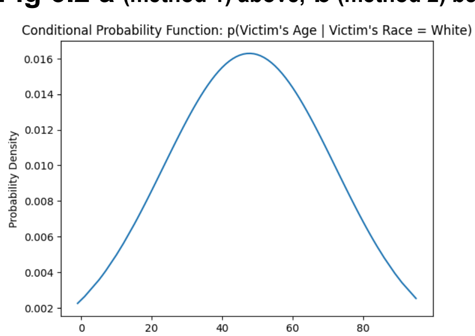


Fig 6.2 a (method 1) above, b (method 2) below



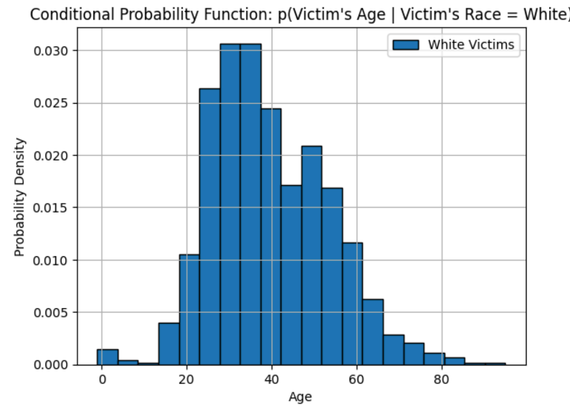
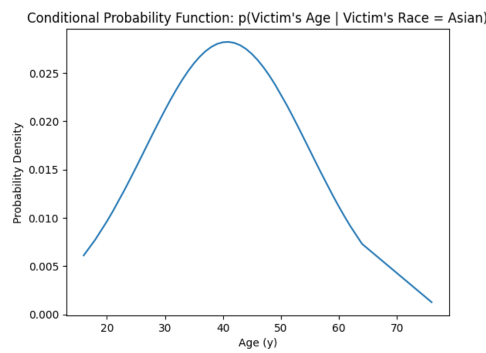


Fig 6.3 a (method 1) above, b (method 2) below



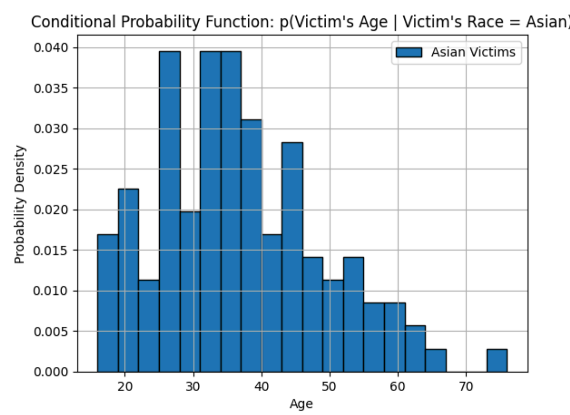
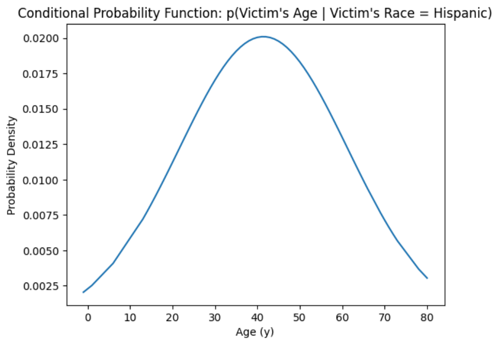
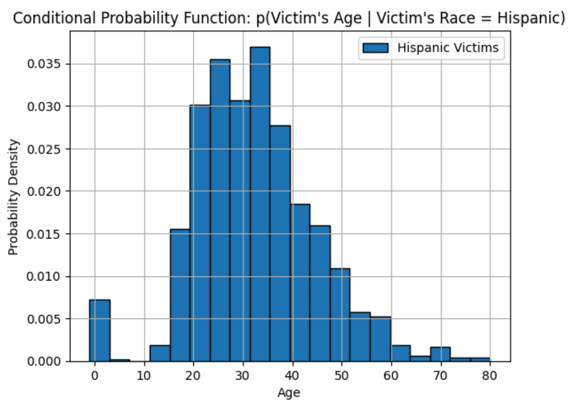


Fig 6.4 a (method 1) above, b (method 2) below





Both methods a and b were included in the report for Fig 6.1-6.4, as while method a makes more sense conceptually, method b makes more sense graphically.

References

[1] E205 Lab 1, https://sites.google.com/g.hmc.edu/e205-fa23/lectures (accessed Sep. 6, 2023).

[2] E80 Lab 7 Navigation, https://sites.google.com/g.hmc.edu/e80/labs/lab-7-navigation (accessed Sep. 6, 2023).

[3] VKS_E205hw1.ipynb, Google Colab, https://colab.research.google.com/drive/1cOIapLjxUzXsIqsL5x6nqpDsAB58SEps?usp=sharing (accessed Sep. 7, 2023).